

RFI — Edge compute, network & barn infrastructure (BSV EquiCare)

To: Edge-computing / systems-integration partner

From: Bharat Sports Venture — EquiCare engineering

Covers: cross-cutting compute, network, power, enclosure, installation and fleet management for all 12 monitoring points

0. How to reply — please read first

Please answer all of this in ONE consolidated reply. We are evaluating several suppliers in parallel on a fixed timeline and will not be sending rounds of follow-up questions. In practice that means:

1. **Answer every numbered question**, in the reply slots provided. If something is not supported, **write that plainly** — "not supported" is a perfectly good answer and costs you nothing. A blank or skipped item, however, we have to record as *not supported*, because we cannot tell the difference.
2. **Attach the documents and files listed in the checklist at the end.** Please send them with this reply, not "on request" — a claim we cannot verify counts as unverified.
3. **Send the actual artifact, not a description of it.** Our single worst experience with a previous supplier: they confirmed "Linux SDK — yes", and the package that arrived was x86-64 only, with no ARM64 build available at all. That cost both sides weeks. So where we ask for a binary, driver, register map or protocol spec, please attach the real file (or a download link) — or state clearly that it does not exist.
4. If a question does not apply to your product, write **"N/A"** and one line saying why.

We would rather receive an honest reply with several "not supported" answers than an optimistic one that unravels at integration.

1. What we are building, and what we are NOT asking you to build

A 24/7 monitoring system for stabled horses in India. Per stall: one dual thermal+optical camera, one IMU leg tag, feed and water sensing, and a microphone. Per barn: an **edge computer**, PoE network, 4G failover and UPS.

Please read this before quoting — it defines the boundary.

BSV has already built, and is not procuring:

- the **camera driver** (thermal ROI control, temperature polling, RTSP) — the camera exposes an HTTP/JSON API (ISAPI) plus Modbus/TCP, so this is done in Python and runs natively on ARM64. **No native SDK wrapper is required.** (An earlier version of this specification asked for one; that is now obsolete.)
- the **edge agent** — sensor ingestion, offline queue, backfill
- the **data contract** — every reading is `{horseId, metric, value, unit, ts, source, confidence}`

- the **backend and rollup engine** — clinical rules, per-horse learned baselines, alerting, monitoring-gap detection
- the **web application** — dashboard, per-horse detail, alerts, reports
- all **CV and DSP models** — activity, vices, gait, urination/excretion, thermal respiration

What we need from you is the hardware platform it runs on, the barn infrastructure around it, and installation/commissioning. If you also offer software services, say so separately — but please do not price the items listed above.

2. Hard requirements — please confirm these first

Pass/fail. If any cannot be met, say so plainly rather than leaving it blank.

#	Requirement	Why
H1	NVIDIA Jetson (ARM64) based edge platform , Ubuntu 22.04+ / current JetPack.	Our entire stack is built and tested for aarch64.
H2	All processing and storage on-premise in India ; cloud sync (if any) only to an India region (e.g. GCP asia-south1). No data transiting or processing outside India — including OTA and telemetry.	DPDP-2023. Please provide a data-flow diagram naming every egress point and its region.
H3	Full local operation during WAN/4G loss — inference, alerting and buffering continue with no internet.	Rural barns lose connectivity routinely.
H4	Barn-rated enclosure — dust, ammonia, humidity, 45–50 °C , insects/rodents, vibration.	Indian barn conditions; a standard office mini-PC enclosure will not survive.

REPLY (H1–H4)

3. Compute sizing — the central question

- Per stall the load is:** up to 3 × RTSP 1080p @25 fps (IR / visible / fusion) decoded on NVDEC, thermal ROI temperature polling at 5–20 Hz over Modbus/HTTP, ROI repositioning over HTTP, one BLE IMU at 100–200 Hz raw, one audio channel at ≥16 kHz — plus **our** concurrent CV (activity, vices, urination/excretion posture, gait) and thermal-respiration DSP.

How many stalls per single edge box at full frame rate, and which exact Jetson tier (Orin Nano / Orin NX 8–16 GB / AGX Orin 32–64 GB) for **1, 4, 8, 16 and 24 stalls**? State GPU TFLOPs, CPU cores and RAM for each.

REPLY

2. **Measured, not theoretical.** Please provide a **benchmark** showing GPU %, NVDEC %, CPU %, RAM and thermal headroom while decoding 3 × 1080p @25 fps per camera and running concurrent inference. State the model(s) used for the benchmark so we can compare against our own.

REPLY

3. **De-rating at temperature.** How many stalls per box at a **45 °C ambient** versus 25 °C, accounting for thermal throttling? This determines our real per-barn count.

REPLY

4. **Storage** — NVMe capacity and endurance per box, with **power-loss protection**. Sizing guidance for derived readings + event clips + audit thermal frames at N stalls, plus retention/rotation. (*The camera itself has only 12 GB, so the edge box carries retention.*)

REPLY

4. Network

5. **PoE switch** specification per barn — per-port and total power budget with **≥25 % headroom**, gigabit uplinks, VLAN/QoS for RTSP — for 4 / 8 / 16 / 24 cameras at ≤6 W each, with runs up to 100 m. Note mics and possibly feed/water sensors may also draw PoE.

REPLY

6. **Bandwidth.** Per-stall and aggregate LAN load (3 × 1080p @25 fps, H.264 and H.265, plus thermal, IMU and audio) — confirm switch backplane and edge NIC are adequate. Then the **WAN/cloud-sync** steady and peak Mbps *after* edge filtering; does that fit typical Indian rural broadband?

REPLY

7. **4G/LTE failover** — dual-SIM industrial router (Indian carriers and bands), failover detection time, and expected **monthly cellular GB per site** after edge filtering. Can it be throttled to readings-only to cap cost during an outage?

REPLY

8. **UPS sizing** — VA/Wh for a stated ride-through, **clean storage-safe shutdown** (no database corruption), automatic restart and state recovery. State total continuous wattage per barn at N stalls.

REPLY

5. Physical install and commissioning

9. **Enclosure** — IP rating, operating temperature, cooling approach (fanless vs filtered), and protection against insects/rodents. Where is it mounted relative to the stalls?

REPLY

10. **Camera mounting** — we need the horse's head in frame at **~3.5 m** with a fixed-focus thermal lens whose in-focus band is roughly **3.0–4.3 m**. Can you survey and mount to hold that geometry, and provide **impact-protected or recessed mounts** (horses will contact the camera; the camera has no published IK rating)?

REPLY

11. **Thermal↔visible alignment**. The camera vendor does **not** expose a visible→thermal coordinate mapping, and reports the fusion overlay as approximate. We therefore detect on the thermal/fusion

stream ourselves. Do you offer any **per-install alignment or calibration tooling** — and if so, time per stall and what happens after a knock?

REPLY

12. **Zero-touch commissioning** — how is a new site brought up: stall→camera→horse mapping, network config, credentials? What is the per-stall install and commissioning time?

REPLY

6. Ingestion of each modality

13. **IMU** — BLE gateway hardware, tags per box/gateway at 100–200 Hz raw, realistic range across a metal/masonry barn, and reconnect behaviour when a horse returns from turnout.

REPLY

14. **Audio** — which interface do you recommend into the Jetson (PoE network mic / USB / I2S)? Include the audio channel in the per-box stall count in Q1.

REPLY

15. **Feed/water** — RS-485/Modbus, 4–20 mA, pulse and BLE inputs available per box; how many devices per bus; and any gateway needed.

REPLY

7. Fleet, resilience and security

16. **Health and failure detection at the platform level** — dead camera, dropped BLE gateway, offline mic, full disk, throttling SoC, **stale clock**. *Our software already flags stale or missing data and refuses to report a blind stall as healthy; we want the hardware layer to surface faults too.*

REPLY

17. **Time discipline** — how is the box's clock kept accurate (NTP source, PTP grandmaster, GPS?), and can it act as an **NTP/PTP server for the cameras and sensors** on the barn LAN? *All cross-sensor fusion depends on this.*

REPLY

18. **Fleet management / OTA across sites** — platform (Balena / Mender / other), OS and container updates, staged or canary rollout with **automatic rollback**, health telemetry, remote reboot, and recovery of a bricked box **without a site visit**.

REPLY

19. **Edge security** — full-disk encryption, secure boot, TPM or secure-element key storage, per-device certificates. Nothing recoverable in plaintext if a box is stolen from a barn.

REPLY

20. **Multi-tenant isolation** — we will run multiple client sites. Per-site credential and key isolation, such that a stolen or compromised box cannot read or write another site's data.

REPLY

8. Deployment model and boundary

21. **Deployment unit** — per barn, or per N stalls? Provide a **per-site bill of materials** (edge box + switch + router + UPS + enclosure + cabling + mounts) at **8, 16 and 24 stalls**.

REPLY

22. **Integration boundary and support** — state precisely what you deliver versus what BSV builds (see §1), the documented handover interface, per-stream acceptance tests, **RMA turnaround in India**, and your support SLA for remote barn sites.

REPLY

9. Evaluation

We would like **one edge unit configured for the pilot** (with switch, router and UPS recommendations), so we can deploy our existing stack on it and measure real stall capacity before scaling.

REPLY (AVAILABILITY + LEAD TIME)

Response checklist

Please confirm each item is attached or answered, so we can evaluate in a single pass:

	Item
☐	Every numbered question answered (or marked "not supported" / "N/A")
☐	All hard requirements (H-numbers) explicitly confirmed or declined
☐	Sensor / component datasheets
☐	Protocol document, register map or API reference — the actual file
☐	ARM64 (aarch64) build attached or download link — or a plain statement that none exists

	Item
☐	Sample data payload showing real field names and units
☐	Sample code, if you provide any
☐	Evaluation-unit availability and lead time
☐	A named technical contact we can reach directly with any single clarification

If anything on this list genuinely cannot be shared before an order, please say which item and why, so we can factor it in rather than assume the worst.

Commercial terms are not part of this RFI — we will discuss those separately once the technical fit is confirmed. Note that our software stack is already built and running; we are looking for the platform and infrastructure to deploy it on, not a software rebuild.